

Usability Evaluation of a VR French Learning App for Google Cardboard

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Abstract: *The purpose of this research work is to investigate cost-effective immersive virtual reality (VR) for learning French language by designing, developing and evaluating Mission Maison, an affordable immersive VR application for Google Cardboard. The development was guided by Design Science Research principle and aligned with John Biggs' constructive alignment theory to transform passive learning into engaging adventures through immersive environments, compelling stories and gamified quizzes that is accessible via smartphone. 29 Malaysian young adults participated in the usability and acceptance evaluation using the System Usability Scale (SUS) in a controlled environment. All the participants were briefed on the application's functionality and went through a tutorial on using Google Cardboard prior to using the VR application. The evaluation revealed a high usability and learnability of Mission Maison, with an average SUS score of 86.38. The evaluation results also indicated that Mission Maison was easy to use, well-integrated, engaging and with minimal learning curves, serving as strong support that cost-effective immersive VR has the potential of springboarding traditional language learning to provide enjoyable and engaging learning experiences. The research work contributes in terms of a new practical budget-friendly immersive VR French language learning application and insights into usability and acceptance by young adults in Malaysia, for broader accessibility and adoption in language education. The findings provide educators, policymakers, and technology developers with a scalable solution to integrate immersive VR instructional design into curricula with aims to enhance language learning experiences.*

Keywords: Instructional Design, Virtual Reality, French Language Learning, Usability Evaluation, Immersive Learning

1. Introduction

In today's global society, language acquisition is important for effective communication. However, engaging young adults with traditional language learning methods is challenging as current language learning methodologies present numerous limitations. For example, learning in a university setting, the traditional methods like memorization of vocabulary lists and repetitive grammar exercises fail to engage students, resulting in low participation and retention rates. Traditional passive ways of teaching, which often do not require active students' participation may not be attractive and adequate to capture tech-savvy young adults. In contrast, teaching with interactive digital platforms like Kahoot! or Quizlet, which incorporates real-time quizzes, multimedia content, and instant feedback, students are more engaged and motivated. This is because the younger generations have been raised in an era of smartphones, social media, and instant access to information. As a result, they are expecting more interactive

and engaging learning experiences. The shift from passive to active learning better aligns with the expectations and learning habits of tech-savvy young adults. For example, classroom-based instructional teaching mode lacks interactivity to engage learners (Wang & Wen, 2023), leading researchers to seek solutions that address such shortfalls. Thus, there is a need to transform language education to address the gaps.

Consequently, researchers have explored whether mobile phone apps could aid language acquisition by offering convenient access for learning anytime and anyplace (Iwata et al., 2023). While widely used smartphone language learning applications such as Duolingo and Babbel are easy to access and well-known, they frequently lack interactive and captivating elements necessary to maintain young adults' interest over an extended period. Their drawbacks have been well documented (Fan et al., 2023). Mobile learning applications often rely on monotonous, repetitive tasks for language acquisition. Moreover, mobile learning applications do not offer the immersive and interactive experiences required for in-depth learning and long-term retention of a new language. Despite the importance of active learner engagement for language acquisition as reported by Hiver et al. (2024) and Alqarni (2023), immersive language learning applications are not commonly available.

Researchers are actively exploring the potential of AR and VR technologies to create immersive and engaging learning experiences aimed at overcoming challenges in experiential learning (Shaukat, 2023). AR technology enriches learning experience by overlaying digital information onto the real environment to create an enhanced and interactive learning space (Daniel, 2023). For language learning, this means pointing a smartphone at objects in real environment and the vocabulary of the objects will appear on the screen. This approach certainly adds an engaging layer to traditional learning methods. However, this way of AR learning relies on the learner's existing environment. VR technology pushes further the boundaries of immersion supporting a transformative leap in the learning experience. Sun et al. (2023) highlight VR's potential to increase engagement and motivation in the educational experience through heightened presence. VR provides learners with a sense of being in the virtual world that AR cannot match. In immersive VR environments, learners are completely transported into a virtual world, leading to increased focus, emotional engagement, and motivation to interact with the virtual environment. The effectiveness of VR outperforms traditional word-word paired association methods in second language vocabulary acquisition (Legault et al, 2019).

The potential of VR in education, especially for language learning, is huge. However, its widespread adoption has been hampered by financial constraint. High-end immersive VR systems, such as Meta Quest or HTC Vive are often expensive and require powerful computers to operate. The high acquisition cost prohibits many educational institutions, especially resource-constrained environments or developing countries to adopt VR technology into education settings. Given such backdrop, an immersive VR French Language Learning Application, known as Mission Maison, was developed in this research work to bridge the gap of immersive VR and its practical implementation in French language education. Mission Maison leverages on the affordable Google Cardboard platform to transform smartphone into VR devices and deliver immersive VR French language learning experiences. The affordable price of Google Cardboard opens the possibility of immersive VR-enhanced language learning to a much wider audience. Mission Maison incorporated the immersive power of VR with storytelling, interactive scenarios, and gamified quizzes as key features to enhance language learning. Instead of presenting vocabulary and phrases in isolation as found in existing mobile

language learning applications, Mission Maison transports learners into a virtual French apartment, supporting contextual learning that mimics real-world language acquisition. For example, instead of matching the word "canapé" to "sofa" on a screen, learners identify the sofa in the virtual 3D living room in Mission Maison.

Although cost-effective VR application has the potential to boost interest and enhance the effectiveness of learning the French language, there are substantial uncertainties surrounding the usability and acceptance of the proposed solution. Therefore, this research work also aims to fill the void with empirical evidence assessing responses from 29 Malaysian young adults, through System Usability Scale (SUS). By gathering and understanding the feedback from young adults' perceptions of Mission Maison's usability and acceptability of acquiring fundamental French language, the study could lead to discoveries that allow for better affordable immersive learning experiences. This research is important because it integrated cost-effective immersive VR technology in French language education. The research findings provide valuable insights into instructional design, technology-enhanced learning, and curriculum innovation. The results could help educators and policymakers understand how cost-effective immersive VR can improve language acquisition and engagement. The remaining part of the paper is structured as follows: Section 2 gives an overview of existing innovative educational technologies that enhance language learning. Section 3 provides details of the design and implementation of Mission Maison. Section 4 discusses the evaluation methodology and presents the evaluation results. The final section concludes the findings and proposes future directions for the research.

2. Related Work

This section discusses various research works that examine the integration of innovative educational technologies to enhance language learning. These technologies improve learning outcomes and accessibility by providing personalized learning experiences, interactive practice, and automated assessment (Major et al., 2021). Over the years, educational technologies have made remarkable progress, evolving from early-stage mobile applications to smartphone applications that utilize AR, and advancing to the use of immersive VR technologies. Each technological leap has contributed to language learning while accompanied by its own limitations in educational settings. It is important to understand the benefits and drawbacks of such educational technologies for effective adoption in language learning and enhance student learning experiences.

Mobile applications allow students to learn conveniently through their mobile devices anytime and anywhere (Iwata et al., 2023). Flexible learning opportunities provided by mobile applications allow students to learn in between classes, during their commute, while waiting for friends, etc., breaking the traditional way of learning in classroom environments. Popular commercial mobile applications to learn languages globally include Duolingo and Babbel (Kessler et al., 2023). Duolingo uses a gamified approach to reward and motivate learners for self-directed language learning experiences (Li, Zixi, and Curtis J. Bonk, 2023). A meta-analysis of 62 studies has shown that mobile learning has a large effect on students' learning outcomes, with smartphones yielding the best results (Garzón et al., 2023). Short microlearning lessons via mobile devices have also shown positive results in improving listening, speaking, writing, and reading skills (Kohnke, 2023). Despite these benefits, mobile learning applications often rely on repetitive tasks for language acquisition, which frequently fails to sustain young adults' interest over time. Recent studies (Fan, Liu, Wang, & Yu, 2023) have reported limited

interaction and engagement in mobile learning. The lack of dynamic and engaging mobile learning experience may lead to diminished motivation and reduced long-term effectiveness in language learning.

To overcome these limitations, researchers overlay learning content information onto the physical environment through the processing power and camera of smartphone. The combination of real and virtual elements, known as AR, has shown promising results in enhancing educational experiences (Shaukat, 2023). Through AR, students acquire language skills in a more immersive learning context coupled with informative interactions to increase motivation and reduce anxiety (Shadiev & Liang, 2024). For instance, students can point their smartphone at various objects in their surroundings and the AR app will overlay vocabulary of these objects in the target language, making language acquisition more engaging and contextually meaningful. Existing studies on AR have been reported to enhance learners' listening skills, grammar learning, and other aspects of language learning (Min & Yu, 2023; Marrahi-Gomez & Belda-Medina, 2023). However, AR technology is unable to completely immerse and contextualize students within the learning environments, leading to a reduction in the effectiveness of AR applications in language education (Lee et al., 2023). Additionally, educators' insufficient knowledge and confidence in using AR technologies restrict the widespread adoption and effective implementation of these tools in language education (Khoshnevisan & Park, 2021).

Meanwhile, the use of immersive VR has also been reported in language education. Through VR, immersive learning environments were created, in which learners could interact with the digital learning objects, leading to better understanding and engagement (Ma et al., 2023). VR can provide learners with learning experiences that relate to real-life contexts. This can boost student engagement and motivation and thus, improve their language skills and competence (Dooly et al., 2023; Huang et al., 2023). VR has demonstrated its effectiveness in helping learners acquire and retain Chinese as a second language, providing an authentic and engaging context that promotes active involvement and language skill development (Chen, & Yuan 2023). In learning English for specific purposes, like geography, VR-based learning has shown improved results compared to traditional video-based methods, particularly in incidental vocabulary acquisition and cognitive, behavioral, and social engagement (Li et al., 2022). These findings highlight VR's potential to enhance language education through immersive and interactive learning experiences.

However, to adopt VR in language education, it requires specific equipment such as head-mounted devices and high-performance computers to render complex virtual environments smoothly. The challenge limits accessibility of VR technology, particularly in resource-constrained educational settings (Cowie & Alizadeh, 2022). Although a more cost-effective immersive VR solution like Google Cardboard exists, there is a need to investigate its usability and acceptance for language learning. Although the benefits of immersive VR for language learning are well-documented, there is a notable scarcity of research on cost-effective solutions such as Google Cardboard. There is a need for the acquisition of empirical evidence to validate the effectiveness and user acceptance of affordable immersive VR technologies across various educational settings. Filling this gap is important for enhancing the practical application of immersive VR in language education, ultimately making it more accessible and impactful. Table 1 provides a summary of advantages and limitations of adopting mobile applications, AR and VR for language learning.

Table 1: Comparison among Mobile Application, AR and VR for language learning

Technology	Advantages	Limitations
Mobile Application	Convenience, portability, anytime/anywhere learning, microlearning for improving language skills (Iwata et al., 2023; Garzón et al., 2023; Kohnke, 2023).	Limited interactivity and engagement have been reported (Fan et al., 2023).
Augmented Reality	Immersive learning context, increased motivation, facilitated interaction, reduced anxiety, effective for enhancing listening skills, grammar learning (Shadiev & Liang, 2024; Min & Yu, 2023; Marrahi-Gomez & Belda-Medina, 2023).	Difficult-to-use interfaces, hindering learning experience (Pereira et al., 2020). Unable to completely immerse students (Lee et al., 2023).
Virtual Reality	Immersive experiences, contextualized learning, engagement, motivation, improved linguistic knowledge and competence (Ma et al., 2023; Dooly et al., 2023; Huang et al., 2023).	Specialized equipment (VR headsets, high-performance computing devices), accessibility limitations in resource-constrained settings (Cowie & Alizadeh, 2022).

3. Design and Development of Mission Maison

The development of Mission Maison is the outcome of a collaboration project between MMU and Education Department of the Embassy of France in Malaysia to revolutionize French language acquisition through educational technologies, in particular, affordable immersive VR technology. Design Science Research (DSR) principle (Absari et al., 2022) is used as the guiding framework for the research work of solving real-world educational problems in language learning. The research begins with an extensive literature review on educational technologies for language learning ranging from mobile applications to AR and VR technologies. The findings reveal the strengths and limitations of various educational innovations and often fall short in terms of providing immersive and interactive learning experiences to learners. Using these insights, researchers decided to harness the strength of immersive VR to develop Mission Maison with key features of immersive learning environments, interactive scenarios, quizzes and narratives to provide learners with engaging French language learning experience. Besides, to ensure high-quality immersive learning experiences are accessible to most learners, Mission Maison will be deployed on Google Cardboard, an affordable VR platform. This will solve the accessibility problems associated with expensive VR equipment.

During the design phase, a French language expert from the Embassy of France in Malaysia was consulted on the educational content and instructional design of Mission Maison. This includes vocabulary selection, pronunciation guides and interactive quizzes in the VR application. The purpose is to ensure the content was suitable and the instructional strategies employed were effective for learning French. Working in parallel, VR designer from MMU

was consulted to ensure visual designs and interactive elements for Mission Maison are intuitive for learners. The role of VR designer's is to develop visually appealing and user-friendly interface that enhances the overall user experience of Mission Maison.

John Biggs' constructive alignment theory (Biggs, 2003) was adopted to design and develop Mission Maison so that the teaching activities and assessments in the engaging and immersive environment are purposefully align with the intended learning outcomes, which are:

1. Mastering French household vocabulary through interactive and contextualized learning experiences.
2. Improving pronunciation and listening comprehension skills via authentic audio guides and interactive tasks.
3. Enhancing engagement and motivation in language learning through immersive VR environments and gamified elements.

The development phase started with choosing a suitable head-mounted display, follow with the creation of immersive virtual learning environment, design of 3D models, the incorporation of non-player characters (NPCs) to enrich narratives, the integration of interactive components, and the design of user-friendly interfaces aimed at ensuring a seamless user experience. Once the initial development phase has been completed, Mission Maison was evaluated by the language expert and a research team to obtain feedback. After the Mission Maison prototype refinement process was completed, a user evaluation was conducted with a targeted group of young adult learners to assess Mission Maison's usability and acceptance. The empirical results from this evaluation aimed to elucidate how cost-effective immersive VR technology, as exemplified by Mission Maison, can positively reshape language learning experiences and outcomes for young adult learners.

Figure 1 summarizes the development process of Mission Maison, from developing applications assets to importing them into Unity and finally deploying the immersive VR application optimizing for Google Cardboard. The focus of the Mission Maison is the creation of an immersive educational environment that capitalizes on the capabilities of VR to provide an engaging and interactive French language learning experience. At the heart of the application assets consists of an array of 3D models, Graphical User Interface (GUI), NPCs, audio clips, etc. The array of 3D models has dual purposes, firstly is to create believable and relatable virtual environments. Items commonly found in a house ranging from elegant furniture to decorations and interactive objects, such as sofa, lamp, TV, curtain etc., are modelled with Blender and Maya, to give a sense of familiarity and context to the French language learning scenarios. The second function is to act as interactive learning objects. For example, when learners pick up a virtual interactive learning object, they can hear the French pronunciation of it. Acquiring vocabulary in this contextual learning approach might be more natural and memorable. GUI elements were developed and integrated into Mission Maison to provide learners with clear instructions, narrations, feedback, and navigation. The GUI serves as visual guide to ensure learners are not lost in virtual home environments. To bring life and make Mission Maison more interesting, NPCs such as wizard and cousin were modelled using Maya and animated with Adobe Mixamo. The NPCs function as virtual conversational partners to engage learners in the simulated home environments with activities and enhance acquisition of the French language in a narratively compelling manner. For instance, upon completion learning of French vocabulary in an area, the wizard will challenge learners to a scavenger hunt via interactive dialogue, asking them to find objects around the house in French upon learning vocabulary in an area. The in-app virtual conversational partners of supporting storytelling

aspect of the Mission Maison and gamification approach aims to enrich narrative and interactive dialogue opportunities, reinforcing French language learning.

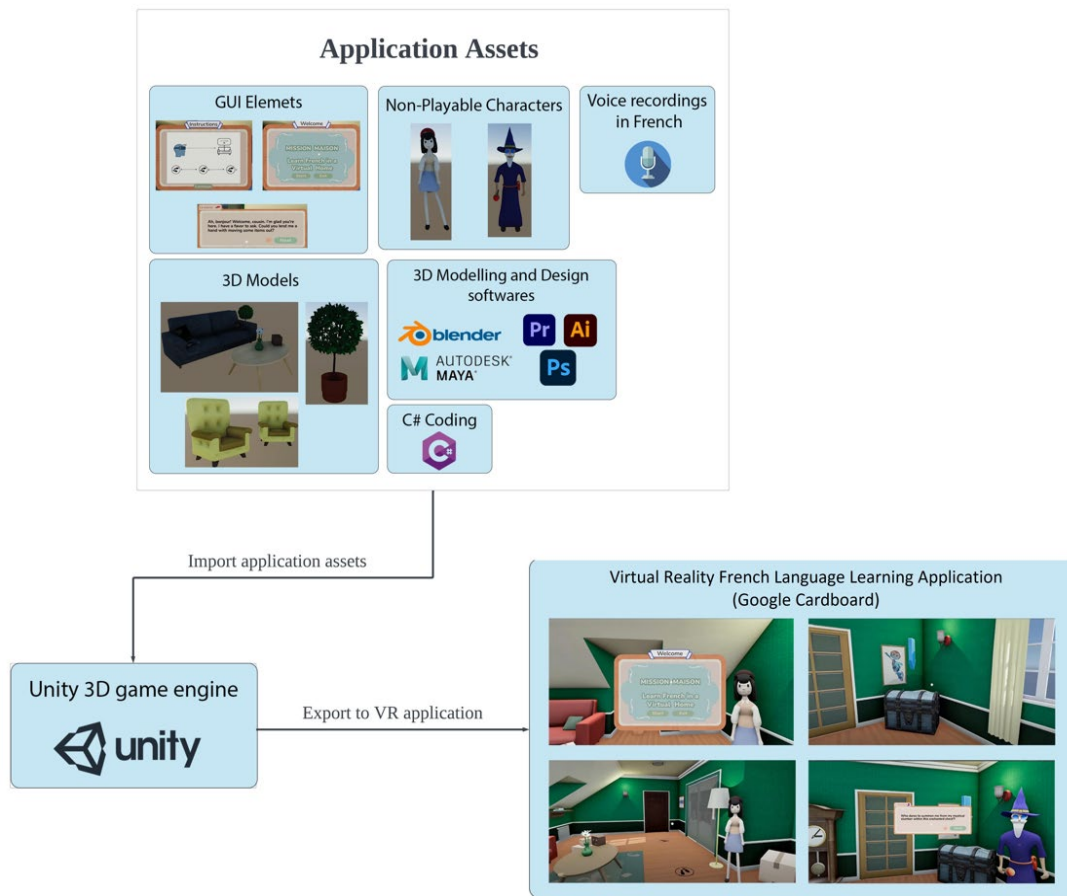


Figure 1: Overall Development Process of Mission Maison

C# codes were scripted to facilitate real-time responses based on learner's progress and interactions. For instance, C# scripts are used to track whether learners correctly locate a piece of furniture in the virtual surroundings based on the given French vocabulary. If learners answer incorrectly, they will be notified and requested to try again. The C# codes enable interactions between various application assets to work as designed to ensure a seamless and immersive user experience. The scripts are attached to the NPCs and interactive objects so that it can react dynamically to learner actions. The purpose is to provide learners with adaptive learning experience so that they can learn at their own pace and style. An important aspect of learning French vocabulary is pronunciation. To provide learners with authentic pronunciation guides and auditory cues, audio recordings from native French speakers were incorporated in Mission Maison. For example, when learners interact with learning objects, they will hear the correct pronunciations. The objective is to enhance learners' pronunciation and listening skills. Once all the application assets were developed, they were imported into Unity game engine to create a cohesive and interactive immersive virtual French learning home environment. Navigation within Mission Maison is achieved through a teleportation mechanic, marked by distinct circular symbols throughout the virtual home environments, as depicted in Figure 2. The circular symbols are placed strategically within the virtual home environment to allow learners to navigate efficiently while maintaining the immersive feel. For instance, when learners explore the virtual living room, circular teleportation symbols are placed near the 'canapé' (sofa). This allows learners to quickly move from one area to another, facilitating

smooth navigation and interaction with different household items. Using teleportation, learners move around the home environments by focusing their gaze on desired locations and activating the teleport function. This navigation approach was chosen to reduce the risk of motion sickness which is a common concern when using VR applications. Finally, Mission Maison was exported to the Google Cardboard platform. The platform was chosen to make immersive language learning experiences widely accessible and affordable, allowing anyone with a smartphone to start experiencing immersive VR-based language learning.



Figure 2: Teleportation navigation within Mission Maison

Educational Content and Assessment

The Mission Maison is specifically developed to assist learners in mastering French household vocabulary with emphasis on vocabulary, pronunciation, and listening comprehension skills. The educational content within the Mission Maison, focused on learning scenarios, where learners immerse themselves in a French home to learn vocabulary related to household items. Through contextual interactions within the virtual environment, learners can retain and recall the vocabulary. For example, learners must interact with different household objects in the living room to learn the vocabulary. This approach of memorizing the vocabulary allows learners to acquire language skills in a fun and engaging way. This approach is aligned with the learning outcome of mastering French household vocabulary via contextual and interactive learning experience. To master French vocabulary words in Mission Maison, a variety of interactive tasks like clicking, memorizing, identifying, choosing, and moving objects were designed to engage learners and enhance their learning experience. Interactive quizzes are embedded throughout the different sections of the virtual apartment to challenge learners to identify and interact with specific household items based on auditory or visual prompts. For example, a wizard is asking the learner to find "un tableau" (a painting), testing the learner vocabulary retention in a practical and contextual manner, as depicted in Figure 3. Learner is requested to identify different pieces of furniture among virtual items scattered throughout the virtual apartment, thus demonstrating the learner comprehension of French vocabulary. These interactive quizzes function as immediate assessments to reinforce learning which is align with the learning outcome of enhancing engagement and motivation in language learning. Besides, the Mission Maison also assesses learners' listening comprehension skills, as illustrated in Figure 4. In this scenario, a native French speaker's voice recording plays, providing learner

with an auditory description of a specific piece of furniture or household item within the virtual space. The learner's task is to carefully listen to the audio cue and visually identify the correct object by gazing at it from the surrounding virtual home environment. The hands-on, visual approach assessment not only evaluates the learners' understanding but also enhances the ability to recognize and associate French words with real-world objects, fostering a more meaningful and contextual learning experience within the Mission Maison. The experiential learning couples with diverse tasks ensures learners remain engaged and motivated throughout the French language learning journey.



Figure 3: Mission Maison Interactive Vocabulary Quiz



Figure 4: Mission Maison Interactive Listening Quiz

Walkthrough of Mission Maison

Upon launching Mission Maison application on a smartphone and donning a Google Cardboard headset, learner is instantly transported into an immersive French apartment. The learning journey begins with a living room of the French apartment as shown in Figure 5. The mirror authentic immersive virtual French home will engage learners in way that traditional textbooks or mobile apps cannot deliver. As the learner navigates around, the softly grown interactive objects placed strategically around the home like "canapé" (sofa), "lampe" (lamp), "un tableau" (a painting), etc., will invite learner to explore further. By gazing at the interactive objects, a GUI will be triggered to visually display the French term and an audio pronunciation by a native speaker. This dual-mode presentation helps reinforce vocabulary acquisition by catering

to both visual and auditory learning styles. The immersive nature of Mission Maison offers unique benefits that set it apart from conventional language learning approaches. The ability to visually see the 3D objects while learning their French names creates powerful mental associations. This level of immersion is not achievable with textbooks or 2D mobile apps. After mastering the vocabulary in a section of the apartment, the learner is prompted to complete interactive quizzes. These quizzes ensure continuous engagement and reinforce learning. These include identifying objects based on their French vocabulary or participating in mini games. The mini games, as depicted in Figure 6, serve as a dynamic method to assess and reinforce vocabulary learning. In the mini game, NPC requests the learner to retrieve a specific recipe as part of a task to prepare a sandwich, requiring the learner to demonstrate understanding of the vocabulary. The gamified elements and progress-based unlocking of new sections of the home could maintain learners' motivation in learning French language. By the end of the immersive virtual apartment tour, learner is expected to acquire basic understanding of French vocabulary related to household items, not only theoretical but also practical and contextual. The combination of visual, auditory, and interactive elements within the immersive Mission Maison creates a memorable and engaging language learning experience that fosters knowledge retention and motivation. This makes affordable immersive VR a powerful tool to engage learners in a way that traditional methods often fail to achieve.



Figure 5: Immersive French Apartment



Figure 6: Mini game: Demonstrating Learner's Understanding of Vocabulary

4. Evaluation Results

A preliminary evaluation was conducted, employing SUS, developed by Brooke (Brooke, 2013), to elicit valuable insights on the Mission Maison's usability and acceptance among young Malaysian adults. SUS is a well-established questionnaire, comprising of 10 items with a 5-point Likert scale ranging from 1 (indicating strong disagreement) to 5 (indicating strong agreement). SUS has gained broad acceptance in multiple fields, including VR applications (Cheong et al., 2023; Daling et al., 2023), because it provides a quick and straightforward way to assess usability even with relatively small sample sizes. SUS scores are calibrated such that they range from 0, which indicates the lowest level of usability, to 100, which signifies the highest level of usability, thereby providing a quantifiable metric of usability as it is perceived by users (Klug in 2017). The SUS evaluation took place at the VR research lab at MMU, in Cyberjaya Malaysia with the purpose of investigating the acceptance of cost-effective immersive VR in French language learning among young Malaysian adults. A convenience sampling method was utilized to recruit 29 young adults, all under the age of 26, for the study. The participants volunteered for the study and received no compensation. The selection of Malaysian as the subject of this study aims to enhance the diversity of data representation, instead of exploring dynamics specific to Malaysia or to make assertions about distinctive Malaysian characteristics.

The evaluation begins with a briefing by a research assistant to each of the participants on the objective, functionalities of Mission Maison and the testing procedures. Then, the participants were requested to provide their consent through Google Forms before starting the evaluation. A tutorial on handling Google Cardboard was given to each participant to ensure proficiency in using the Mission Maison. An immersive French language acquisition with the Mission Maison within a fully operational French home environment was facilitated, with each participant given an approximately 25 minutes to experience it. Throughout the evaluation, the participants were guided to navigate and experience the Mission Maison. After experiencing the Mission Maison, the participants were requested to fill in 10-item SUS questionnaire using Google Form and continued with a question-and-answer session to obtain additional insights from the participants.

The SUS evaluation revealed a positive usability and learnability perceptions among young Malaysian adults towards the Mission Maison as the average SUS score was 86.38 with a median of 87.5. A summary of the SUS scores by the 29 participants is shown in Figure 7. The average SUS score surpasses the threshold of 85.5 for excellent usability as suggested by Bangor et al. (2009), and the low standard deviation of 5.45 signifies consistent user experiences across the participants. The high SUS scores imply that participants found the Mission Maison easy to learn and use. The seamless blend of immersive VR environments, storytelling narratives, and interactive language learning scenarios within the Mission Maison seems to have resonated well with the participants, fostering an engaging and intuitive learning experience for young Malaysian adults to acquire basic French language skills. The minimum SUS score was 70, although it is lower than the average, it remains within the acceptable usability range, suggesting that even users with relatively lower perceptions did not find the Mission Maison unusable. Conversely, the maximum score of 100 from a participant perceived the Mission Maison as flawless in terms of usability.

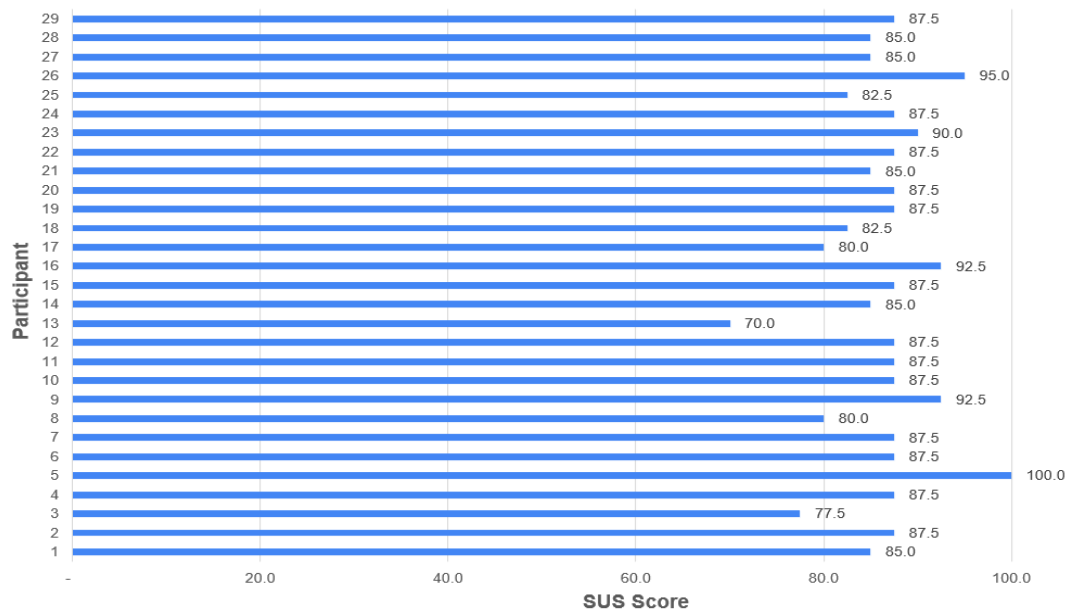


Figure 7: SUS scores from 29 participants

A thorough examination of the individual SUS question responses was analyzed as shown in Table 2, to obtain valuable insights into the specific usability and learnability dimensions of the Mission Maison. The purpose is to understand its potential acceptance and effectiveness as an immersive French language learning tool. Most of the participants would like to use the Mission Maison frequently as shown in the outcome from Q1 ($M=4.31$, $Md=4.00$). This implies that the utilization of cost-effective immersive VR has a high potential to keep the engagement and interest of the participants over time. This is important as existing educational technology such as smartphone language learning applications are not able to engage young adults over an extended period. The participants also perceived the Mission Maison to be free from excessive complexity, and easy to use. This is evidenced by the scores of Q2 ($M=1.59$, $Md=2.00$) and Q3 ($M=4.34$, $Md=4.00$). This means immersive VR technology does not become a barrier to the participants. This is crucial as it allows learners to focus their attention and mental efforts towards acquiring and retaining the French language content. The participants were able to use the Mission Maison without much assistance as depicted in the score of Q4 ($M=1.66$, $Md=2.00$) which means the participants learnt to operate the application quickly. This could promote autonomous learning and empower learners. Moving on to Q5 ($M=4.55$, $Md=5.00$), it illuminates the participants' favorable perception of the integration of the Mission Maison's functionalities. The seamless transition between different interactive learning scenarios and uninterrupted exploration of the learning environments might have contributed to the cohesive user experience. Shifting to score of Q6 ($M=1.38$, $MD=1.00$), it reflects minimal inconsistencies in interface and functionality of the Mission Maison. Followed by scores of Q7 ($M=4.48$, $MD=5.00$) and Q8 ($M=1.52$, $MD=1.00$), suggesting that the learning curve to operate the Mission Maison was minimal, and it was easy to comprehend by the participants. Response of Q9 ($M=4.55$, $Md=5.00$), the participants were very confident in using the Mission Maison, which is an important factor for learner motivation, engagement, and autonomy. Results for Q10 ($M=1.55$, $Md=1.00$) suggest no substantial initial learning was required. However, the SD of 0.74 indicates there is variability in the participants' experiences, which could be influenced by their diverse prior experience with immersive VR. In short, the positive participant perceptions across dimensions such as ease of use, functional integration, and rapid learning curve suggest that the Mission Maison has the potential to enhance learner

engagement and facilitate knowledge retention, thereby addressing the challenges of traditional classroom-based language learning approaches.

Table 2: Evaluation Results for Individuals SUS Question

SUS Question		Mean (M)	Median (Md)	Min	Max	Standard Deviation (SD)
Q1	I think that I would like to use this Mission Maison frequently.	4.31	4.00	2.00	5.00	0.71
Q2	I found this Mission Maison unnecessarily complex.	1.59	2.00	1.00	3.00	0.57
Q3	I thought this Mission Maison was easy to use.	4.34	4.00	3.00	5.00	0.67
Q4	I think that I would need assistance to be able to use this Mission Maison.	1.66	2.00	1.00	3.00	0.72
Q5	I found the various functions in this Mission Maison were well integrated.	4.55	5.00	3.00	5.00	0.57
Q6	I thought there was too much inconsistency in this Mission Maison.	1.38	1.00	1.00	2.00	0.49
Q7	I would imagine that most people would learn to use this Mission Maison very quickly.	4.48	5.00	3.00	5.00	0.57
Q8	I found this Mission Maison very cumbersome/awkward to use.	1.52	1.00	1.00	3.00	0.57
Q9	I felt very confident using this Mission Maison.	4.55	5.00	4.00	5.00	0.51
Q10	I needed to learn a lot of things before I could get going with this Mission Maison.	1.55	1.00	1.00	4.00	0.74

Based on the evaluation results, participants found Mission Maison to be easy to use, well-integrated, and quick to learn. This suggests that it will boost students' engagement and improve knowledge retention. The implementation of a cost-effective immersive VR application through Mission Maison addresses the shortcomings of traditional language learning approaches. Firstly, from passive reception of information in traditional methods to active participation in VR. For instance, instead of passively reading vocabulary lists, learners actively participate in the compelling stories, navigate around the virtual apartment and interact with learning objects to learn French vocabulary and pronunciation, which enhances their engagement and retention of the language. Through gamification elements, learning becomes more enjoyable and can engage students more deeply with the learning content. For example, learners must correctly identify objects based on the vocabulary they have learned in a virtual scavenger hunt to unlock new stories and advance to the next level. This engaging approach makes the learning process exciting and motivates them to practice more frequently. Second, from theoretical learning in a typical classroom-based learning to practice language skills in simulated immersive real-life contexts. The immersive and interactive nature of VR engages

students with learning content in a hands-on manner to foster better knowledge retention and understanding. This is difficult to achieve with traditional methods. Third, instead of monotonous learning through conventional approaches, VR captivates students' attention through dynamic and interactive elements in immersive environments. Through immersive VR, students are fully immersed in contextual and practical language settings, thereby making the learning experience distinctly relevant and relatable. Overall, incorporating immersive VR environments, storytelling, quizzes, and interactive scenarios into language curriculums has huge potential to revolutionize the way young adults learn. Besides, the implementation of cost-effective immersive VR applications is financially sustainable for educational institutions. Thus, allowing broad access without significant budget increases. Mission Maison can serve as a model for implementing affordable and accessible VR applications in education, allowing more institutions to benefit from the advantages of immersive learning. Furthermore, immersive VR applications can be scaled across multiple classrooms or institutions to ensure widespread adoption and utilization. This study contributes empirical insights of usability of cost-effective immersive VR application to French language learning. However, it is essential to acknowledge the limitations of our study, including the need for larger sample sizes and diverse demographics to enhance generalizability. Subsequence study can focus on accessing long-term usability and acceptance of cost-effective immersive VR language learning applications. Additionally, further research should explore longitudinal studies to assess the enduring effectiveness of cost-effective immersive VR language learning applications.

5. Conclusion

The empirical evaluation of the Mission Maison for Google Cardboard has unveiled promising results for language acquisition among young adult learners. The high average SUS score and favorable score for individual SUS items of the Mission Maison among young Malaysian adults indicate the application's usability and acceptance of cost-effective immersive VR to address the existing challenges associated with traditional language learning approaches. Affordability of Google Cardboard eliminates the challenge of acquiring specialized VR devices when adopting immersive VR into language education. Consequently, broaden the use of immersive VR technology in resource-limited institutions. The integration of immersive virtual learning environment, interactive scenarios, and engaging narratives within the Mission Maison has demonstrated its usefulness in enhancing learner engagement through active participation and knowledge retention. This study demonstrates the potential of cost-effective immersive VR through implementation of Mission Maison to improve French language acquisition. By transporting learners into a contextualized French apartment setting, they are able to learn vocabulary and pronunciations skills in a practical language application in daily life. Immersive VR promotes active interaction between learners and the learning content, transforming passive learning into engaging practical experience. Through immersive VR contextual learning scenarios, it can facilitate a better retention and understanding of French vocabulary. Mission Maison provides a personalized learning experience, allowing learners to learn at their own pace and speed. This boosts learners' confidence and keeps them engage in learning French language. through various VR-based educational activities. In short, this research work has demonstrated the potential of cost-effective immersive VR to transform traditional language learning methods through experiential learning, engagement, and learner autonomy. This study contributes to the field of educational technology and language learning by emphasizing the importance of economical immersive VR to address the limitations of traditional methods. Cost-effective immersive VR applications offer great potential to revolutionize language education, foster more engaging and effective learning experiences, and ultimately enhance

language proficiency among learners. Although Mission Maison was positively accepted, it is imperative to evaluate the long-term impact on language proficiency and the potential of integration into formal education. Subsequent studies can prioritize exploring the application of VR in teaching various languages to different age groups. Future research could focus on qualitative investigations that may yield valuable insights into specific design elements or user experiences that contribute to the overall positive perceptions.

Acknowledgement

The authors wish to extend their gratitude for the financial support provided by Education Department of the Embassy of France in Malaysia, MMUE/220009. We also like to express our profound appreciation to Anaïs Deschamps for her exceptional contribution in devising the structure of the game, French language expert and for her invaluable guidance throughout the implementation of the VR application and extend our sincere gratitude to Julien Bigot for his invaluable assistance with the voice recordings for the Mission Maison. We are also grateful to MMU for the provision of VR facilities, which were important in the successful implementation of the Mission Maison.

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